

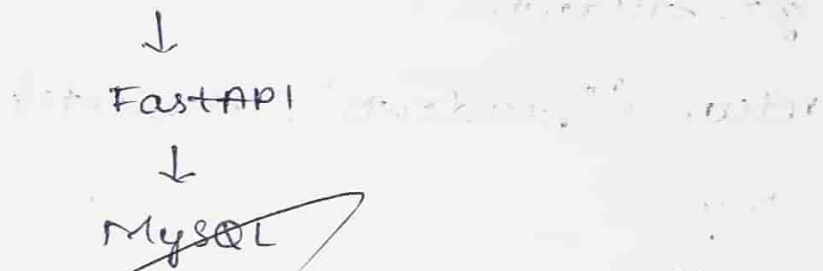
Week-5 FastAPI CRUD Operations

(Skill & Practical)

1. What is FastAPI

It is a python framework used to create APIs. API allows a client (Swagger, frontend, ...) to communicate with a backend.

Basic flow: Client / Swagger



2. CRUD Operations

Operation	HTTP method	Meaning
Create	POST	Add new data
Read	GET	Get data
Update	PUT	Change existing data
Delete	DELETE	Remove data

3. Basic FastAPI Setup

```
from fastapi import FastAPI
```

```
app = FastAPI()
```

app is the FastAPI application

4. GET - Read Data

Purpose

Used to retrieve / get data.

Basic Syntax

```
@app.get("/students")
```

```
def get_students():
```

```
    return {"students": students}
```

With MySQL

```
@app.get("/students")
```

```
def get_students():
```

```
    with engine.connect() as connection:
```

```
        result = connection.execute(  
            text("SELECT * FROM students")  
        )
```

```
        students = result.mappings().all()
```

```
    return {"students": students}
```

5. POST - Create Data

Purpose

Used to add new data.

First define the data format:

```
class Student(BaseModel):
```

```
    id: int
```

```
    name: str
```

Get

↓

Select

↓

Read data

Post

↓

Insert

↓

Add data

department: str

Then:

```
@app.post("/students")
```

```
def create_student(student: Student):
```

```
    with engine.begin() as connection:
```

```
        connection.execute(
```

```
            text("""
```

```
                INSERT INTO students (id, name,
                                      department)
```

```
                VALUES (:id, :name, :department)
```

```
            """),
```

```
        {
```

```
            "id": student.id,
```

```
            "name": student.name,
```

```
            "department": student.department
```

```
        }
```

```
    )
```

```
    return {"message": "Student created"}
```

6. PUT - Update Data

Purpose

Used to change existing data.

{student_id} means the ID comes from the URL.

Put

↓

Update ... Where ID

↓

change data

```
→ @app.put("/students/{student_id}")
```

```
def update_student(student_id: int, student:
```

```
    Student):
```

with engine.begin() as connection:

connection.execute(

text("""

UPDATE students

SET name = :name,

department = :department

WHERE id = :student-id

{

"student-id": student-id,

"name": student.name,

"department": student.department

}

)

~~return~~

return {"message": "Student updated"}

Example:

PUT /students/103

means:

Find student whose ID = 103 and update their details.

7. DELETE - Remove Data

Delete

Purpose

Delete from... where

Used to delete existing data

Remove data

→ @app.delete("/students/{student_id}")

```
def delete_student(student_id: int):
```

```
    with engine.begin() as connection:
```

```
        connection.execute(
```

```
            text("""
```

```
                DELETE FROM students
```

```
                WHERE id = :student_id
```

```
            """))
```

```
            {"student_id": student_id}
```

```
        )
```

```
    return {"message": "Student deleted"}
```

Ex:

```
DELETE /students/103
```

means:

Delete the student whose ID = 103

8. MySQL Connection

```
from sqlalchemy import create_engine, text
```

```
from sqlalchemy.engine import URL
```

```
connection_url = URL.create(
```

```
    "mysql+pymysql",
```

```
    username="root",
```

```
    password="PASSWORD",
```

```
    host="localhost",
```


database = "university"

)

engine = create_engine(connection_url)

Why?

FastAPI



SQLAlchemy



PyMySQL



MySQL

10. Swagger

FastAPI automatically provides Swagger UI for testing APIs.

Open:

<http://127.0.0.1:8000/docs>

Can test:

GET

/students

POST

/students

PUT

/students/{student_id}

DELETE

/students/{student_id}

POST body:

{

"id": 101,

"name": "Ravi",

"department": "CSE"

}

POT

URL : /students/101

Body:

{

"id" : 101,

"name" : "Ravi Kumar",

"department" : "ECE"

}

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